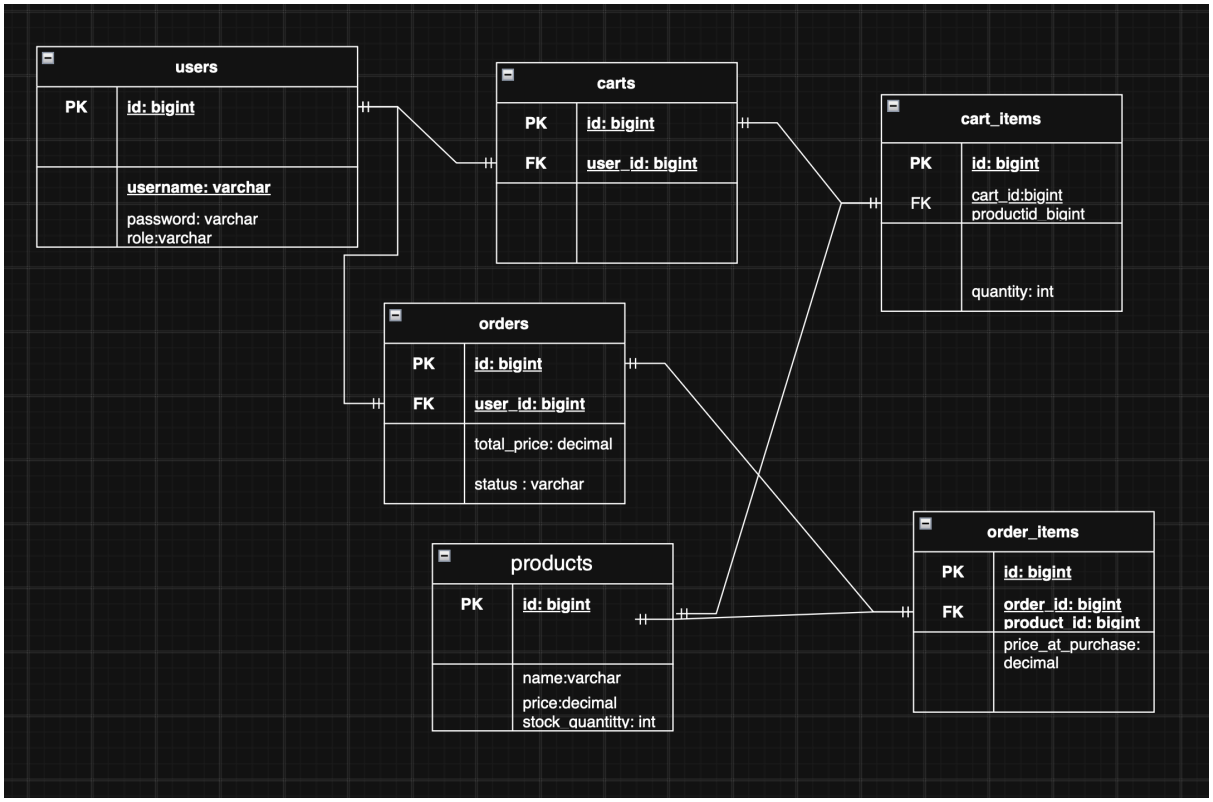
C. Database Design

1. Description of the Main Entities and Relationships The database is designed using a relational model and consists of the following core entities:

- **users:** Stores system users (customers and administrators). It has a one-to-many relationship with `orders` and a one-to-one relationship with `carts`.
- **products:** Represents the store's catalog. It has a one-to-many relationship with both `cart_items` and `order_items`.
- **carts:** Represents an active shopping session linked to a specific user.
- **cart_items:** An associative entity linking `products` to a specific `cart`, tracking the selected quantity.
- **orders:** Represents a completed purchase. It stores the total price, status, and is linked to the user who placed it.
- **order_items:** An associative entity linking `products` to an `order`. It stores the quantity and the exact `price_at_purchase` to maintain historical accuracy even if the main product price changes later.

2. Explanation of How the Data is Stored and Retrieved The system uses a relational database (**PostgreSQL**) for persistent data storage.

- **Storage:** The Business Logic layer maps Java objects to database tables using Spring Data JPA annotations (e.g.,`@Entity`, `@OneToMany`). When a user makes a purchase, the system creates entities and JPA automatically translates these into SQL `INSERT` statements.
- **Retrieval:** Data is accessed through the Data Access Layer using interfaces that extend `JpaRepository`. This allows the application to perform CRUD (Create, Read, Update, Delete) operations and complex table joins without writing raw SQL queries.


users
PK
id: bigint
username: varchar
password: varchar
role:varchar
carts
PK
id: bigint
FK
user_id: bigint
cart_items
PK
id: bigint
FK
cart_id:bigint
productid_bigint
quantity: int
orders
PK
id: bigint
FK
user_id: bigint
total_price: decimal
status : varchar
products
PK
id: bigint
name:varchar
price:decimal
stock_quantitty: int
order_items
PK
id: bigint
FK
order_id: bigint
product_id: bigint
price_at_purchase: decimal